Aabid Mohamed A

BACKEND · DISTRIBUTED SYSTEMS & AI INFRASTRUCTURE

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SUMMARY

Backend engineer with production experience on **agentic workflows and async services** at PwC (enterprise banking client — confidential). Flagship open-source work: voice AI infrastructure (VoxForge), event-driven Go services (BugSathi), and schema-bound LLM extraction (SignalMine). B.Tech, NIT Calicut. Case studies and metrics on brohammad.tech.

EXPERIENCE

PwC — Enterprise banking (confidential) Aug 2025 – Present · Bengaluru

Associate Software Engineer · Backend + AI

- Building an agentic deal platform on **async Python microservices** — document and chat ingestion, NLP extraction, ML classification, and operational APIs for internal users.
- Normalized conversational deal data into structured documents to improve downstream ML extraction quality; added **per-call evaluation logging** to catch extraction drift.
- Shipped classification and blotter APIs on **PostgreSQL** with **Azure** deployment discipline; typed contracts between ingestion, extraction, and scoring stages.

SchoolHack.AI Jan 2025 – Feb 2025 · Remote

Machine Learning Intern

- Multi-source research agent orchestrating web search and enrichment APIs into structured outputs; async FastAPI endpoints.
- Outreach automation prototype with LinkedIn/Gmail integrations.

SELECTED PROJECTS

VoxForge — Voice AI infrastructure 426 collected · 76.40% cov.

Python · FastAPI · LangGraph · LiveKit · pgvector · Docker · OpenTelemetry

- Open-source voice platform — one **VoicePipelineService** for WebSocket, REST, and LiveKit WebRTC; LangGraph agents; pgvector RAG; per-turn eval; signed replay; human handoff.
- Metrics per docs/project-metrics.md: 407 non-browser tests passed, Playwright CI, one-command HTTPS deploy. Case study: brohammad.tech/projects/voxforge

BugSathi — Event-driven bug reporting 116 tests · 35 ADRs

Go · Kafka · PostgreSQL · MinIO · Redis · ffmpeg · OpenTelemetry · Kubernetes

- Ordered **Kafka pipeline** from MinIO upload through ffmpeg frames to AI-drafted reports; API + worker binaries, JWT refresh rotation, OTel/Prometheus/Grafana.
- **116 tests** under **-race** in CI. Case study: brohammad.tech/projects/bugsathi

SignalMine — Market intelligence pipeline 125 tests · 4 schemas

Python · FastAPI · PostgreSQL · Pydantic · Instructor · OpenTelemetry · Prometheus

- GitHub/Reddit collectors → deterministic normalization → schema-validated LLM extraction; golden-dataset field-level evaluation.
- **125 pytest tests** (see docs/project-metrics.md). Case study: brohammad.tech/projects/signalmine

PaintCRM — Dealer platform Live demo

Node.js · Express · PostgreSQL · Docker · Playwright

- Offline-first room preview and CRM workflows; live at paintcrm.brohammad.tech. Case study: brohammad.tech/projects/paint-intelligence

TECHNICAL SKILLS

Languages: Python, Go, TypeScript, JavaScript

Backend: FastAPI, Express, REST, gRPC, WebSockets, Kafka, Redis, async services

AI / platforms: LangGraph, RAG, MCP, LLM evaluation, Instructor, voice AI (STT/TTS/LiveKit)

Data & ops: PostgreSQL, pgvector, MinIO, Docker, Kubernetes, GitHub Actions, Prometheus, Grafana, OpenTelemetry, pytest, Playwright

EDUCATION

National Institute of Technology Calicut B.Tech in Electrical & Electronics Engineering · 2021 – 2025

LEADERSHIP

- **Joint Secretary, Enquire (NIT Calicut):** Statewide outreach for Samasya science competition.
- **IEEE NIT Calicut:** Large-scale student events; branch recognized nationally.